

# LINUX COMMANDS

## CHAPTER 1 : SYSTEM INFORMATION COMMANDS

1) uname:

```
m_sangeethallal@cloudshell:~$ uname
Linux
```

2) uptime:

```
m_sangeethallal@cloudshell:~$ uptime
07:29:19 up 2:40, 0 user, load average: 0.31, 0.64, 0.37
```

3) hostname:

```
m_sangeethallal@cloudshell:~$ hostname
aiserver
```

4) free:

```
m_sangeethallal@cloudshell:~$ free
              total        used         free       shared    buff/cache   available
Mem:      8138088      2358144      4474108          1144       1556916       5779944
Swap:              0              0              0
```

5) top:

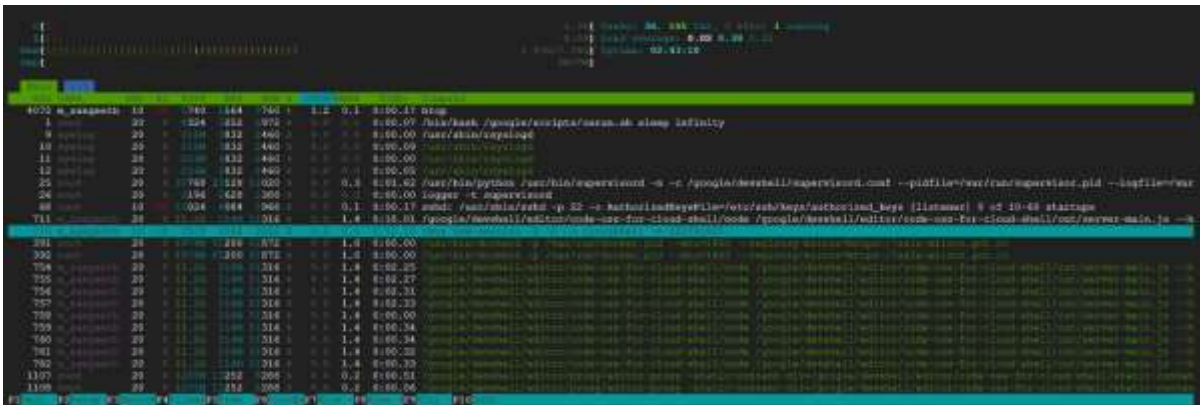
```
m_sangeethallal@cloudshell:~$ top
```

```
%Cpu(s):  0.7 us,  0.5 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.2 si,  0.2 st
MiB Mem : 7947.4 total, 4367.0 free, 2305.0 used, 1520.5 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 5642.3 avail Mem
```

| PID  | USER     | PR | NI  | VIRT    | RES    | SHR   | S | %CPU | %MEM | TIME+   | COMMAND      |
|------|----------|----|-----|---------|--------|-------|---|------|------|---------|--------------|
| 1520 | m_sange+ | 20 | 0   | 1341972 | 79724  | 48384 | S | 0.3  | 1.0  | 0:08.95 | node         |
| 1    | root     | 20 | 0   | 4324    | 3252   | 2972  | S | 0.0  | 0.0  | 0:00.08 | bash         |
| 9    | syslog   | 20 | 0   | 222140  | 3828   | 2460  | S | 0.0  | 0.0  | 0:00.13 | rsyslogd     |
| 25   | root     | 20 | 0   | 38768   | 28128  | 11020 | S | 0.0  | 0.3  | 0:01.60 | python       |
| 26   | root     | 20 | 0   | 6196    | 2628   | 2388  | S | 0.0  | 0.0  | 0:00.00 | logger       |
| 68   | root     | 10 | -10 | 12024   | 4084   | 2968  | S | 0.0  | 0.1  | 0:00.16 | sshd         |
| 329  | root     | 20 | 0   | 2021836 | 85200  | 61872 | S | 0.0  | 1.0  | 0:01.24 | dockerd      |
| 414  | root     | 20 | 0   | 1234268 | 13380  | 8288  | S | 0.0  | 0.2  | 0:01.85 | editor-proxy |
| 415  | root     | 20 | 0   | 16328   | 6416   | 5492  | S | 0.0  | 0.1  | 0:00.01 | sudo         |
| 419  | root     | 20 | 0   | 1226100 | 2700   | 1976  | S | 0.0  | 0.0  | 0:00.01 | tmux-agent   |
| 454  | root     | 20 | 0   | 1793224 | 45552  | 32184 | S | 0.0  | 0.6  | 0:08.07 | containerd   |
| 646  | root     | 20 | 0   | 2696    | 1532   | 1440  | S | 0.0  | 0.0  | 0:00.00 | sleep        |
| 684  | root     | 20 | 0   | 9196    | 4128   | 3752  | S | 0.0  | 0.1  | 0:00.00 | runuser      |
| 689  | m_sange+ | 20 | 0   | 4324    | 3336   | 3068  | S | 0.0  | 0.0  | 0:00.00 | sh           |
| 711  | m_sange+ | 20 | 0   | 11.3g   | 117272 | 51316 | S | 0.0  | 1.4  | 0:28.51 | node         |
| 716  | m_sange+ | 10 | -10 | 7508    | 3032   | 2032  | S | 0.0  | 0.0  | 0:00.35 | tmux         |
| 717  | m_sange+ | 10 | -10 | 8800    | 5884   | 4016  | S | 0.0  | 0.1  | 0:00.24 | bash         |
| 1566 | m_sange+ | 20 | 0   | 8924    | 6052   | 4064  | S | 0.0  | 0.1  | 0:00.04 | bash         |
| 2894 | m_sange+ | 20 | 0   | 11.0g   | 87072  | 49788 | S | 0.0  | 1.1  | 0:01.29 | node         |
| 2985 | m_sange+ | 20 | 0   | 9020    | 6076   | 4056  | S | 0.0  | 0.1  | 0:00.08 | bash         |
| 3441 | root     | 10 | -10 | 13996   | 9784   | 8408  | S | 0.0  | 0.1  | 0:00.03 | sshd         |
| 3443 | root     | 10 | -10 | 13996   | 9628   | 8248  | S | 0.0  | 0.1  | 0:00.02 | sshd         |
| 3445 | m_sange+ | 10 | -10 | 14256   | 6376   | 4720  | S | 0.0  | 0.1  | 0:00.00 | sshd         |
| 3446 | m_sange+ | 10 | -10 | 14256   | 6280   | 4620  | S | 0.0  | 0.1  | 0:00.03 | sshd         |
| 3448 | m_sange+ | 10 | -10 | 7340    | 3780   | 3532  | S | 0.0  | 0.0  | 0:00.00 | bash         |
| 3451 | m_sange+ | 10 | -10 | 7604    | 4344   | 3736  | S | 0.0  | 0.1  | 0:00.00 | bash         |

6) http:

```
m_sangeethallal@cloudshell:~$ http
```



7) arch:

```
m_sangeethallal@cloudshell:~$ arch
x86_64
```

## CHAPTER 2: FILE AND DIRECTORY MANAGEMENT

1) cd:

```
m_sangeethallal@cloudshell:~$ cd students
m_sangeethallal@cloudshell:~/students$
```

2) mkdir:

```
m_sangeethallal@cloudshell:~$ mkdir School
m_sangeethallal@cloudshell:~$
```

3) ls:

```
m_sangeethallal@cloudshell:~$ ls
README-cloudshell.txt  School  students  teachers
```

4) pwd:

```
m_sangeethallal@cloudshell:~$ pwd
/home/m_sangeethallal
```

5) rmdir:

```
m_sangeethallal@cloudshell:~$ rmdir students
m_sangeethallal@cloudshell:~$
```

6) touch:

```
m_sangeethallal@cloudshell:~$ cd School
m_sangeethallal@cloudshell:~/School$ touch students_data.txt
m_sangeethallal@cloudshell:~/School$ ls
students_data.txt
```

7) rm:

```
m_sangeethallal@cloudshell:~/School$ rm students_data.txt
m_sangeethallal@cloudshell:~/School$ ls
m_sangeethallal@cloudshell:~/School$
```

8) cp:

```
m_sangeethallal@cloudshell:~/School$ touch file1.txt
m_sangeethallal@cloudshell:~/School$ touch file2.txt
m_sangeethallal@cloudshell:~/School$ cp file1.txt file2.txt
m_sangeethallal@cloudshell:~/School$
```

9) mv:

```
m_sangeethallal@cloudshell:~/School$ mv file1.txt notes.txt
m_sangeethallal@cloudshell:~/School$
```

10) ln:

```
m_sangeethallal@cloudshell:~/School$ touch students_data.txt
m_sangeethallal@cloudshell:~/School$ ln students_data.txt link1
m_sangeethallal@cloudshell:~/School$ ln -s students_data.txt link2
m_sangeethallal@cloudshell:~/School$ ls -l
total 0
-rw-rw-r-- 1 m_sangeethallal m_sangeethallal 0 May 14 09:44 file2.txt
-rw-rw-r-- 2 m_sangeethallal m_sangeethallal 0 May 14 10:12 link1
lrwxrwxrwx 1 m_sangeethallal m_sangeethallal 17 May 14 10:12 link2 -> students_data.txt
-rw-rw-r-- 1 m_sangeethallal m_sangeethallal 0 May 14 09:44 notes.txt
-rw-rw-r-- 2 m_sangeethallal m_sangeethallal 0 May 14 10:12 students_data.txt
```

## CHAPTER 3: FILE VIEWING AND EDITING

1) cat:

```
m_sangeethallal@cloudshell:~/School$ echo "Students data 2026" > students_data.txt
m_sangeethallal@cloudshell:~/School$ cat students_data.txt
Students data 2026
```

2) more:

```
m_sangeethallal@cloudshell:~/School$ truncate -s 0 students_data.txt
m_sangeethallal@cloudshell:~/School$ echo "Students data file" > students_data.txt
m_sangeethallal@cloudshell:~/School$ more students_data.txt
Students data file
```

3) less:

```
Students data file
Students data file
```

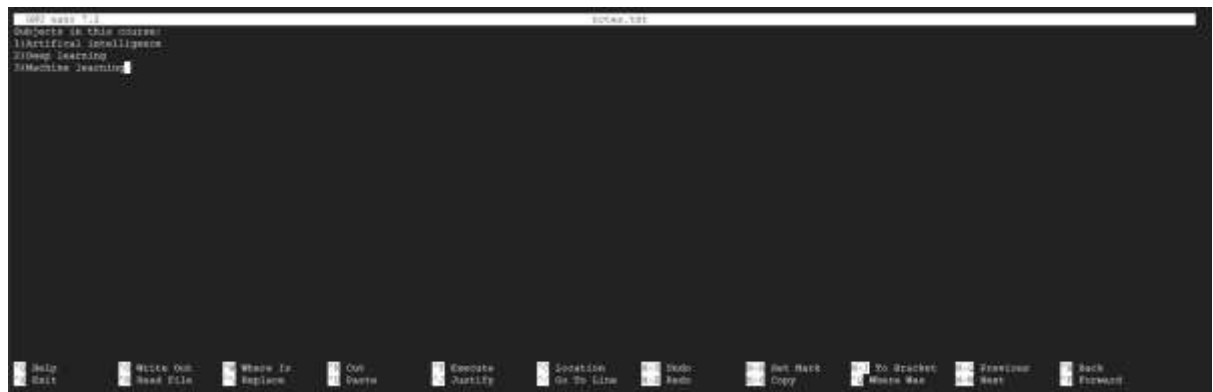
4) tail:

```
m_sangeethallal@cloudshell:~/School$ cat students_data.txt
class 12 students
Rahul, CSE, 2026
Priya, ECE, 2026
m_sangeethallal@cloudshell:~/School$ tail students_data.txt
class 12 students
Rahul, CSE, 2026
Priya, ECE, 2026
m_sangeethallal@cloudshell:~/School$ tail -1 students_data.txt
Priya, ECE, 2026
```

5) head:

```
m_sangeetha1a1@cloudshell:~/School$ head students_data.txt
class 12 students
Rahul, CSE, 2026
Priya, ECE, 2026
m_sangeetha1a1@cloudshell:~/School$ head -1 students_data.txt
class 12 students
```

6) nano:



7) vi/vim:



8) awk:

```
m_sangeetha1a1@cloudshell:~/School$ awk '{print $1}' students_data.txt
class
Rahul,
Priya,
```

9) sed:

```
m_sangeetha1a1@cloudshell:~/School$ sed 's/Rahul/Amit/' students_data.txt
class 12 students
Amit, CSE, 2026
Priya, ECE, 2026
```

## CHAPTER 4: FILE PERMISSIONS AND OWNERSHIP

1) chmod:

```
m_sangeetha1a1@cloudshell:~/School$ chmod o+r students_data.txt
m_sangeetha1a1@cloudshell:~/School$ ls -l students_data.txt
-rwxrwxrwx 2 m_sangeetha1a1 m_sangeetha1a1 52 May 14 10:29 students_data.txt
```

2) chown:



```
m_sangeethallal@cloudshell:~/School$ sudo useradd student1
m_sangeethallal@cloudshell:~/School$ chown student1 students_data.txt
chown: changing ownership of 'students_data.txt': Operation not permitted
m_sangeethallal@cloudshell:~/School$ sudo chown student1 students_data.txt
m_sangeethallal@cloudshell:~/School$ ls -l students_data.txt
-rwxrwxrwx 2 student1 m_sangeethallal 52 May 14 10:29 students_data.txt
```

3) chgrp:

```
m_sangeethallal@cloudshell:~/School$ touch notes.txt
m_sangeethallal@cloudshell:~/School$ sudo groupadd developers
m_sangeethallal@cloudshell:~/School$ sudo chgrp developers notes.txt
m_sangeethallal@cloudshell:~/School$ ls -l notes.txt
-rw-rw-r-- 1 m_sangeethallal developers 85 May 14 14:57 notes.txt
```

4) umask:

```
m_sangeethallal@cloudshell:~/School$ umask 027
m_sangeethallal@cloudshell:~/School$ touch demo.txt
m_sangeethallal@cloudshell:~/School$ ls -l demo.txt
-rw-r----- 1 m_sangeethallal m_sangeethallal 0 May 14 14:59 demo.txt
```

5) stat:

```
m_sangeethallal@cloudshell:~/School$ stat demo.txt
  File: demo.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131525         Links: 1
Access: (0640/-rw-r-----)  Uid: ( 1000/m_sangeethallal)   Gid: ( 1000/m_sangeethallal)
Access: 2026-05-14 14:59:58.181739867 +0000
Modify: 2026-05-14 14:59:58.181739867 +0000
Change: 2026-05-14 14:59:58.181739867 +0000
 Birth: 2026-05-14 14:59:58.181739867 +0000
```

6) lsattr:

```
m_sangeethallal@cloudshell:~/School$ lsattr demo.txt
-----e----- demo.txt
```

7) chattr:

```
m_sangeethallal@cloudshell:~/School$ sudo chattr +i demo.txt
m_sangeethallal@cloudshell:~/School$ lsattr demo.txt
---i-----e----- demo.txt
```

8) usermod:

```
m_sangeethallal@cloudshell:~/School$ sudo useradd riya
useradd: user 'riya' already exists
m_sangeethallal@cloudshell:~/School$ sudo passwd riya
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
m_sangeethallal@cloudshell:~/School$ sudo passwd riya
New password:
Retype new password:
passwd: password updated successfully
m_sangeethallal@cloudshell:~/School$ sudo usermod -aG sudo riya
m_sangeethallal@cloudshell:~/School$ groups rahul
groups: 'rahul': no such user
m_sangeethallal@cloudshell:~/School$ groups riya
riya : riya sudo
m_sangeethallal@cloudshell:~/School$ id riya
uid=1002(riya) gid=1003(riya) groups=1003(riya),27(sudo)
```

## CHAPTER 5: NETWORKING COMMANDS

### 1) ip addr:

```
m_sangeethallal@cloudshell:~/School$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
```

### 2) ip route:

```
m_sangeethallal@cloudshell:~/School$ ip route
default via 10.88.0.1 dev eth0
10.88.0.0/16 dev eth0 proto kernel scope link src 10.88.0.3
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
```

### 3) ping:

```
m_sangeethallal@cloudshell:~/School$ ping google.com
PING google.com (142.251.10.139) 56(84) bytes of data:
64 bytes from sd-in-f139.1e100.net (142.251.10.139): icmp_seq=1 ttl=114 time=0.510 ms
64 bytes from sd-in-f139.1e100.net (142.251.10.139): icmp_seq=2 ttl=114 time=0.292 ms
64 bytes from sd-in-f139.1e100.net (142.251.10.139): icmp_seq=3 ttl=114 time=0.354 ms
64 bytes from sd-in-f139.1e100.net (142.251.10.139): icmp_seq=4 ttl=114 time=0.334 ms
```

### 4) netstat:

```
m_sangeethallal@cloudshell:~/School$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:971             0.0.0.0:*               LISTEN
tcp        0      0 localhost:8998           0.0.0.0:*               LISTEN
tcp        0      0 localhost:34917          0.0.0.0:*               LISTEN
tcp        0      0 localhost:43481          0.0.0.0:*               LISTEN
tcp        0      0 localhost:39341          0.0.0.0:*               LISTEN
tcp        0      0 localhost:36174          localhost:ssh            ESTABLISHED
tcp        0      0 localhost:971            localhost:53722          ESTABLISHED
tcp        0      0 localhost:ssh            localhost:36174          ESTABLISHED
tcp        0      0 localhost:36696          localhost:970            ESTABLISHED
tcp        0      0 localhost:ssh            localhost:36158          ESTABLISHED
tcp        0      0 cs-803328956808-d:51918 si-in-f100.1e100.:https ESTABLISHED
tcp        0      0 localhost:53722          localhost:971            ESTABLISHED
tcp        0      0 localhost:37452          localhost:ssh            TIME_WAIT
tcp        0      0 localhost:53730          localhost:971            ESTABLISHED
tcp        0      0 localhost:36680          localhost:970            ESTABLISHED
tcp        0      0 localhost:36158          localhost:ssh            ESTABLISHED
tcp        0      0 localhost:971            localhost:53730          ESTABLISHED
```

### 5) ss:

```
m_sangeethallal@cloudshell:~/School$ ss -l
Netid State  Recv-Q Send-Q               Local Address:Port
Peer Address:Port                               Process
nl      UNCONN  960    0                *
                                           rtnl:dockerd/331
nl      UNCONN  0      0                *
                                           rtnl:kernel
nl      UNCONN  4352   0                *
                                           tcpdiag:ss/2089
nl      UNCONN  960    0                *
                                           tcpdiag:kernel
nl      UNCONN  0      0                *
                                           xfrm:dockerd/331
```

### 6) wget:

```
m_sangeethallal@cloudshell:~/School$ wget https://example.com/file.zip
--2026-05-14 15:31:41-- https://example.com/file.zip
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10::ac42:93f3, .
Connecting to example.com (example.com)|104.20.23.154|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-14 15:31:41 ERROR 404: Not Found.
```

7) curl:

```
m_sangeethallal@cloudshell:~/School$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
```

8) nslookup:

```
m_sangeethallal@cloudshell:~/School$ nslookup google.com
Server:          169.254.169.254
Address:         169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 74.125.200.101
Name:   google.com
Address: 74.125.200.139
Name:   google.com
Address: 74.125.200.100
Name:   google.com
Address: 74.125.200.113
Name:   google.com
Address: 74.125.200.138
Name:   google.com
Address: 74.125.200.102
Name:   google.com
Address: 2404:6800:4003:c04::71
Name:   google.com
Address: 2404:6800:4003:c04::65
Name:   google.com
Address: 2404:6800:4003:c04::66
Name:   google.com
Address: 2404:6800:4003:c04::8a
```

9) dig:

```
m_sangeethallal@cloudshell:~/School$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<- opcode: QUERY, status: NOERROR, id: 44908
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                277     IN      A      74.125.200.101
google.com.                277     IN      A      74.125.200.139
google.com.                277     IN      A      74.125.200.100
google.com.                277     IN      A      74.125.200.113
google.com.                277     IN      A      74.125.200.138
google.com.                277     IN      A      74.125.200.102

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 15:33:54 UTC 2026
;; MSG SIZE rcvd: 135
```

## CHAPTER 6: PROCESS MANAGEMENT

1) ps:

```
m_sangeetha11a1@cloudshell:~$ ps
  PID TTY          TIME CMD
  693 pts/2    00:00:00 bash
 1502 pts/2    00:00:00 ps
```

2) pgrep:

```
m_sangeetha11a1@cloudshell:~$ pgrep bash
1
671
674
688
693
```

3) pkill:

```
m_sangeetha11a1@cloudshell:~$ sleep 500 &
[1] 1580
m_sangeetha11a1@cloudshell:~$ pgrep sleep
655
1580
m_sangeetha11a1@cloudshell:~$ pkill sleep
pkill: killing pid 655 failed: Operation not permitted
[1]+  Terminated                  sleep 500
m_sangeetha11a1@cloudshell:~$ pgrep sleep
655
```

4) kill:

```
m_sangeetha11a1@cloudshell:~$ sleep 500 &
[1] 1594
m_sangeetha11a1@cloudshell:~$ pgrep sleep
655
1594
m_sangeetha11a1@cloudshell:~$ kill 1594
[1]+  Terminated                  sleep 500
```

5) killall:

```
m_sangeetha11a1@cloudshell:~$ sleep 500 &
[1] 1602
m_sangeetha11a1@cloudshell:~$ sleep 500 &
[2] 1607
m_sangeetha11a1@cloudshell:~$ pgrep sleep
655
1602
1607
m_sangeetha11a1@cloudshell:~$ killall sleep
sleep(655): Operation not permitted
[1]-  Terminated                  sleep 500
[2]+  Terminated                  sleep 500
m_sangeetha11a1@cloudshell:~$ pgrep sleep
655
```

6) bg:

```
m_sangeetha11a1@cloudshell:~$ sleep 500
^Z
[1]+  Stopped                  sleep 500
m_sangeetha11a1@cloudshell:~$ bg
[1]+  sleep 500 &
m_sangeetha11a1@cloudshell:~$ jobs
[1]+  Running                  sleep 500 &
m_sangeetha11a1@cloudshell:~$ fg
sleep 500
^Z
[1]+  Stopped                  sleep 500
```



7) fg:

```
m_sangeetha11a1@cloudshell:~$ sleep 500
^Z
[1]+  Stopped                  sleep 500
m_sangeetha11a1@cloudshell:~$ bg
[1]+  sleep 500 &
m_sangeetha11a1@cloudshell:~$ jobs
[1]+  Running                  sleep 500 &
m_sangeetha11a1@cloudshell:~$ fg
sleep 500
^Z
[1]+  Stopped                  sleep 500
```

8) jobs:

```
m_sangeetha11a1@cloudshell:~$ sleep 500
^Z
[1]+  Stopped                  sleep 500
m_sangeetha11a1@cloudshell:~$ bg
[1]+  sleep 500 &
m_sangeetha11a1@cloudshell:~$ jobs
[1]+  Running                  sleep 500 &
m_sangeetha11a1@cloudshell:~$ fg
sleep 500
^Z
[1]+  Stopped                  sleep 500
```

9) nice:

```
m_sangeetha11a1@cloudshell:~$ nice
-10
```

10) renice:

```
m_sangeetha11a1@cloudshell:~$ sleep 500 &
[2] 1635
m_sangeetha11a1@cloudshell:~$ ps -l
F S  UID      PID      PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000      693      692  0  70 -10 -   2361 do_wai pts/2    00:00:00 bash
0 T  1000     1619      693  0  70 -10 -   1422 do_sig pts/2    00:00:00 sleep
0 S  1000     1635      693  0  70 -10 -   1422 hrtime pts/2    00:00:00 sleep
0 R  1000     1638      693  0  70 -10 -   2730 -      pts/2    00:00:00 ps
m_sangeetha11a1@cloudshell:~$ renice 10 -p 1635
1635 (process ID) old priority -10, new priority 10
```

## CHAPTER 7:DISK MANAGEMENT

1) df:

```
m_sangeetha11a1@cloudshell:~$ df
Filesystem                1K-blocks      Used Available Use% Mounted on
overlay                  98831908 76304264  22511260  78% /
tmpfs                     65536          0     65536   0% /dev
/dev/sda1                 98831908 76304264  22511260  78% /root
/dev/disk/by-id/google-home-part1 5018272  156528  4583268   4% /home
/dev/root                 2019696 1288424   731272  64% /usr/lib/modules
shm                       65536          0     65536   0% /dev/shm
[2]-  Done                  sleep 500
```

2) du:

```
m_sangeetha11a1@cloudshell:~$ du School
16      School
```

### 3) fdisk:

```
m_sangeetha11a1@cloudshell:~$ sudo fdisk -l
Disk /dev/sda: 100 GiB, 107374182400 bytes, 209715200 sectors
Disk model: PersistentDisk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: gpt
Disk identifier: 88CD0047-6E3D-984A-9279-9D30D1B6BC8F

Device            Start       End   Sectors   Size Type
/dev/sda1    8704000 209715166 201011167 95.8G Linux filesystem
/dev/sda2      20480     53247     32768    16M ChromeOS kernel
/dev/sda3   4509696  8703999   4194304    2G ChromeOS root fs
/dev/sda4      53248     86015     32768    16M ChromeOS kernel
/dev/sda5   315392  4509695   4194304    2G ChromeOS root fs
/dev/sda6     16448    16448         1    512B ChromeOS kernel
/dev/sda7     16449    16449         1    512B ChromeOS root fs
/dev/sda8     86016   118783     32768    16M Linux filesystem
/dev/sda9     16450    16450         1    512B ChromeOS reserved
/dev/sda10    16451    16451         1    512B ChromeOS reserved
/dev/sda11         64    16447    16384     8M BIOS boot
/dev/sda12   249856   315391    65536    32M EFI System
```

### 4) mkfs:

```
m_sangeetha11a1@cloudshell:~$ sudo mkfs -t ext4 /dev/sda1
mke2fs 1.47.0 (5-Feb-2023)
/dev/sda1 is mounted; will not make a filesystem here!
```

### 5) mount:

```
m_sangeetha11a1@cloudshell:~$ sudo mount /dev/sdb1 /mnt
m_sangeetha11a1@cloudshell:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0 100G  0 disk
├─sda1       8:1    0  95.8G  0 part /mnt
│           /run/google/devshell
│           /var/lib/docker
│           /etc/ssh/keys
│           /var/config/tmux
│           /etc/resolv.conf
│           /etc/hostname
│           /etc/hosts
│           /root
├─sda2       8:2    0   16M  0 part
├─sda3       8:3    0    2G  0 part
├─sda4       8:4    0   16M  0 part
├─sda5       8:5    0    2G  0 part
├─sda6       8:6    0   512B  0 part
├─sda7       8:7    0   512B  0 part
├─sda8       8:8    0   16M  0 part
├─sda9       8:9    0   512B  0 part
├─sda10      8:10   0   512B  0 part
├─sda11      8:11   0    8M  0 part
└─sda12      8:12   0   32M  0 part
sdb          8:16   0    5G  0 disk
└─sdb1       8:17   0    5G  0 part /mnt
                        /home
```

### 6) unmount:

```
m_sangeetha11a1@cloudshell:~$ sudo umount /dev/sdb1
m_sangeetha11a1@cloudshell:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
├─sda
│   ├─sda1   8:1      0 95.8G  0 part /mnt
│           /run/google/devshell
│           /var/lib/docker
│           /etc/ssh/keys
│           /var/config/tmux
│           /etc/resolv.conf
│           /etc/hostname
│           /etc/hosts
│           /root
│   ├─sda2   8:2      0   16M  0 part
│   ├─sda3   8:3      0    2G  0 part
│   ├─sda4   8:4      0   16M  0 part
│   ├─sda5   8:5      0    2G  0 part
│   ├─sda6   8:6      0  512B  0 part
│   ├─sda7   8:7      0  512B  0 part
│   ├─sda8   8:8      0   16M  0 part
│   ├─sda9   8:9      0  512B  0 part
│   ├─sda10  8:10     0  512B  0 part
│   ├─sda11  8:11     0    8M  0 part
│   └─sda12  8:12     0   32M  0 part
└─sdb
   └─sdb1   8:17     0    5G  0 part /home
```

7) blkid:

```
m_sangeetha11a1@cloudshell:~$ sudo blkid
/dev/sdb1: UUID="b9f0e455-8c03-483b-a0d6-629a47561fb9" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="45a676-01"
/dev/dm-0: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sda4: PARTLABEL="KERN-B" PARTUUID="76f23ba4-cdd3-1143-91d2-9a5f7372c111"
/dev/sda2: PARTLABEL="KERN-A" PARTUUID="2762bea5-16e2-5944-b00a-f19370e6ab65"
/dev/sda12: SEC_TYPE="msdos" LABEL_FATBOOT="EFI-SYSTEM" LABEL="EFI-SYSTEM" UUID="7BE5-F6D0" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI-SYSTEM" PARTUUID="1b9d2508-4204-1a41-81b8-2ddc3bfe9b40"
/dev/sda9: PARTLABEL="reserved" PARTUUID="9634bb25-70d7-1a45-8b09-3648a7de5fe8"
/dev/sda10: PARTLABEL="reserved" PARTUUID="9f5dc8da-5cc5-f44e-8289-f69f69e82b1d"
/dev/sda7: PARTLABEL="ROOT-C" PARTUUID="9bf2b5b6-84ac-a94a-b087-360efbff48b7"
/dev/sda5: PARTLABEL="ROOT-B" PARTUUID="969395aa-4a21-b54f-a5ae-3d373e0d8005"
/dev/sda3: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="ROOT-A" PART
```

8) lsblk:

```
m_sangeetha11a1@cloudshell:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
├─sda
│   ├─sda1   8:1      0 95.8G  0 part /mnt
│           /run/google/devshell
│           /var/lib/docker
│           /etc/ssh/keys
│           /var/config/tmux
│           /etc/resolv.conf
│           /etc/hostname
│           /etc/hosts
│           /root
│   ├─sda2   8:2      0   16M  0 part
│   ├─sda3   8:3      0    2G  0 part
│   ├─sda4   8:4      0   16M  0 part
│   ├─sda5   8:5      0    2G  0 part
│   ├─sda6   8:6      0  512B  0 part
│   └─sda7   8:7      0  512B  0 part
```

9) parted:

```
m_sangeetha11a1@cloudshell:~$ sudo parted /dev/sda
GNU Parted 3.6
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) print
Model: Google PersistentDisk (scsi)
Disk /dev/sda: 107GB
Sector size (logical/physical): 512B/4096B
Partition Table: gpt
Disk Flags: pmbr_boot
```

| Number | Start  | End    | Size   | File system | Name       | Flags                  |
|--------|--------|--------|--------|-------------|------------|------------------------|
| 11     | 32.8kB | 8421kB | 8389kB |             | RWW        | bios_grub              |
| 6      | 8421kB | 8422kB | 512B   |             | KERN-C     | chromeos_kernel        |
| 7      | 8422kB | 8422kB | 512B   |             | ROOT-C     |                        |
| 9      | 8422kB | 8423kB | 512B   |             | reserved   |                        |
| 10     | 8423kB | 8423kB | 512B   |             | reserved   |                        |
| 2      | 10.5MB | 27.3MB | 16.8MB |             | KERN-A     | chromeos_kernel        |
| 4      | 27.3MB | 44.0MB | 16.8MB |             | KERN-B     | chromeos_kernel        |
| 8      | 44.0MB | 60.8MB | 16.8MB | ext4        | OEM        |                        |
| 12     | 128MB  | 161MB  | 33.6MB | fat16       | EFI-SYSTEM | boot, legacy_boot, esp |
| 5      | 161MB  | 2309MB | 2147MB |             | ROOT-B     |                        |
| 3      | 2309MB | 4456MB | 2147MB | ext2        | ROOT-A     |                        |
| 1      | 4456MB | 107GB  | 103GB  | ext4        | STATE      |                        |

10) tune2fs:

```
m_sangeetha11a1@cloudshell:~$ sudo apt install e2fsprogs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
e2fsprogs is already the newest version (1.47.0-2.4~explubuntu4.1).
0 upgraded, 0 newly installed, 0 to remove and 38 not upgraded.
m_sangeetha11a1@cloudshell:~$ sudo tune2fs -l /dev/sda1
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: STATE
Last mounted on: /var
```

## CHAPTER 8: SYSTEM MONITORING AND PERFORMANCE

1) top:

```
%Cpu(s):  0.7 us,  0.5 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.2 si,  0.2 st
MiB Mem :  7947.4 total,  4367.0 free,  2305.0 used,  1520.5 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  5642.3 avail Mem
```

| PID  | USER     | PR | NI  | VIRT    | RES    | SHR   | S | %CPU | %MEM | TIME+   | COMMAND      |
|------|----------|----|-----|---------|--------|-------|---|------|------|---------|--------------|
| 1520 | m_sange+ | 20 | 0   | 1341972 | 79724  | 48384 | S | 0.3  | 1.0  | 0:08.95 | node         |
| 1    | root     | 20 | 0   | 4324    | 3252   | 2972  | S | 0.0  | 0.0  | 0:00.08 | bash         |
| 9    | syslog   | 20 | 0   | 222140  | 3828   | 2460  | S | 0.0  | 0.0  | 0:00.13 | rsyslogd     |
| 25   | root     | 20 | 0   | 38768   | 28128  | 11020 | S | 0.0  | 0.3  | 0:01.60 | python       |
| 26   | root     | 20 | 0   | 6196    | 2628   | 2388  | S | 0.0  | 0.0  | 0:00.00 | logger       |
| 68   | root     | 10 | -10 | 12024   | 4084   | 2968  | S | 0.0  | 0.1  | 0:00.16 | sshd         |
| 329  | root     | 20 | 0   | 2021836 | 85200  | 61872 | S | 0.0  | 1.0  | 0:01.24 | dockerd      |
| 414  | root     | 20 | 0   | 1234268 | 13380  | 8288  | S | 0.0  | 0.2  | 0:01.85 | editor-proxy |
| 415  | root     | 20 | 0   | 16328   | 6416   | 5492  | S | 0.0  | 0.1  | 0:00.01 | sudo         |
| 419  | root     | 20 | 0   | 1226100 | 2700   | 1976  | S | 0.0  | 0.0  | 0:00.01 | tmux-agent   |
| 454  | root     | 20 | 0   | 1793224 | 45552  | 32184 | S | 0.0  | 0.6  | 0:08.07 | containerd   |
| 646  | root     | 20 | 0   | 2696    | 1532   | 1440  | S | 0.0  | 0.0  | 0:00.00 | sleep        |
| 684  | root     | 20 | 0   | 9196    | 4128   | 3752  | S | 0.0  | 0.1  | 0:00.00 | runuser      |
| 689  | m_sange+ | 20 | 0   | 4324    | 3336   | 3068  | S | 0.0  | 0.0  | 0:00.00 | sh           |
| 711  | m_sange+ | 20 | 0   | 11.3g   | 117272 | 51316 | S | 0.0  | 1.4  | 0:28.51 | node         |
| 716  | m_sange+ | 10 | -10 | 7508    | 3032   | 2032  | S | 0.0  | 0.0  | 0:00.35 | tmux         |
| 717  | m_sange+ | 10 | -10 | 8800    | 5884   | 4016  | S | 0.0  | 0.1  | 0:00.24 | bash         |
| 1566 | m_sange+ | 20 | 0   | 8924    | 6052   | 4064  | S | 0.0  | 0.1  | 0:00.04 | bash         |
| 2894 | m_sange+ | 20 | 0   | 11.0g   | 87072  | 49788 | S | 0.0  | 1.1  | 0:01.29 | node         |
| 2985 | m_sange+ | 20 | 0   | 9020    | 6076   | 4056  | S | 0.0  | 0.1  | 0:00.08 | bash         |
| 3441 | root     | 10 | -10 | 13996   | 9784   | 8408  | S | 0.0  | 0.1  | 0:00.03 | sshd         |
| 3443 | root     | 10 | -10 | 13996   | 9628   | 8248  | S | 0.0  | 0.1  | 0:00.02 | sshd         |
| 3445 | m_sange+ | 10 | -10 | 14256   | 6376   | 4720  | S | 0.0  | 0.1  | 0:00.00 | sshd         |
| 3446 | m_sange+ | 10 | -10 | 14256   | 6280   | 4620  | S | 0.0  | 0.1  | 0:00.03 | sshd         |
| 3448 | m_sange+ | 10 | -10 | 7340    | 3780   | 3532  | S | 0.0  | 0.0  | 0:00.00 | bash         |
| 3451 | m_sange+ | 10 | -10 | 7604    | 4344   | 3736  | S | 0.0  | 0.1  | 0:00.00 | bash         |





8) glance:

|                                          |          |        |      |       |       |        |           |        |                  |
|------------------------------------------|----------|--------|------|-------|-------|--------|-----------|--------|------------------|
|                                          |          | CPU ↑  | 1.0% | MEM - | 27.7% | SWAP - | 0.0%      | LOAD - | 2core            |
| CPU                                      | [ 1.0%]  | user   | 3.7% | total | 7.76G | total  | 0         | 1 min  | 0.40             |
| MEM                                      | [ 27.7%] | system | 3.2% | used  | 2.15G | used   | 0         | 5 min  | 0.22             |
| SWAP                                     | [ 0.0%]  | iowait | 0.0% | free  | 5.61G | free   | 0         | 15 min | 0.08             |
| TASKS 33 (124 thr), 1 run, 31 slp, 1 oth |          |        |      |       |       |        |           |        |                  |
| NETWORK                                  |          | Rx/s   | Tx/s |       |       |        |           |        |                  |
| eth0                                     |          | 12Kb   | 24Kb |       |       |        |           |        |                  |
| lo                                       |          | 15Kb   | 15Kb |       |       |        |           |        |                  |
|                                          |          |        |      | CPU%  | MEM%  | PID    | USER      | THR    | NI S             |
| DefaultGateway                           |          |        | 8ms  | >1.0  | 0.6   | 4637   | m_sangeet | 1      | -10 R python3 /u |
|                                          |          |        |      | 0.5   | 0.0   | 692    | m_sangeet | 1      | -10 S tmux new-s |
|                                          |          |        |      | 0.0   | 8.8   | 1722   | m_sangeet | 13     | 0 S node --dns   |
| DISK I/O                                 |          | R/s    | W/s  | 0.0   | 7.6   | 1785   | m_sangeet | 11     | 0 S node /goog   |
| dm-0                                     |          | 0      | 0    | 0.0   | 2.7   | 1771   | m_sangeet | 11     | 0 S node /goog   |
| sda                                      |          | 0      | 309K | 0.0   | 1.4   | 752    | m_sangeet | 11     | 0 S node /goog   |
| sda1                                     |          | 0      | 309K | 0.0   | 1.0   | 312    | root      | 10     | 0 S dockerd -p   |
| sda2                                     |          | 0      | 0    | 0.0   | 0.9   | 1763   | m_sangeet | 7      | 0 S cloudcode_   |
| sda3                                     |          | 0      | 0    | 0.0   | 0.8   | 1733   | m_sangeet | 11     | 0 S node /goog   |
| sda4                                     |          | 0      | 0    | 0.0   | 0.8   | 1756   | m_sangeet | 7      | 0 S node /goog   |
| sda5                                     |          | 0      | 0    | 0.0   | 0.6   | 459    | root      | 8      | 0 S containerd   |
| sda6                                     |          | 0      | 0    | 0.0   | 0.3   | 25     | root      | 1      | 0 S python /us   |

## CHAPTER 9: USER MANAGEMENT

1) useradd:

```
m_sangeetha11a1@cloudshell:~$ sudo useradd rahul
m_sangeetha11a1@cloudshell:~$ sudo passwd rahul
New password:
Retype new password:
passwd: password updated successfully
m_sangeetha11a1@cloudshell:~$ id rahul
uid=1001(rahul) gid=1001(rahul) groups=1001(rahul)
m_sangeetha11a1@cloudshell:~$
```

2) userdel:

```
m_sangeetha11a1@cloudshell:~$ sudo userdel arun
m_sangeetha11a1@cloudshell:~$ id arun
id: 'arun': no such user
m_sangeetha11a1@cloudshell:~$
```

3) usermod:

```
m_sangeetha11a1@cloudshell:~$ sudo usermod -l arun rahul
m_sangeetha11a1@cloudshell:~$ id arun
uid=1001(arun) gid=1001(rahul) groups=1001(rahul)
m_sangeetha11a1@cloudshell:~$
```

4) passwd:

```
m_sangeetha11a1@cloudshell:~$ sudo useradd rahul
m_sangeetha11a1@cloudshell:~$ sudo passwd rahul
New password:
Retype new password:
passwd: password updated successfully
m_sangeetha11a1@cloudshell:~$ id rahul
uid=1001(rahul) gid=1001(rahul) groups=1001(rahul)
m_sangeetha11a1@cloudshell:~$
```

5) chage:

```
m_sangeetha11a1@cloudshell:~$ sudo chage -l arun
Last password change           : May 15, 2026
Password expires                : never
Password inactive               : never
Account expires                 : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
```

#### 6) groups:

```
m_sangeetha11a1@cloudshell:~$ groups arun
arun : rahul
```

#### 7) groupadd:

```
m_sangeetha11a1@cloudshell:~$ sudo groupadd developers
m_sangeetha11a1@cloudshell:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
```

```
pcp:x:990:
glances:x:107:
rahul:x:1001:
developers:x:1002:
```

#### 8) groupdel:

```
m_sangeetha11a1@cloudshell:~$ sudo groupdel developers
m_sangeetha11a1@cloudshell:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,m_sangeetha11a1
```

```
postgres:x:106:
m_sangeetha11a1:x:1000:
pcp:x:990:
glances:x:107:
rahul:x:1001:
```

#### 9) newgrp:

```
m_sangeetha11a1@cloudshell:~$ sudo groupdel developers
m_sangeetha11a1@cloudshell:~$ sudo groupadd developers
m_sangeetha11a1@cloudshell:~$ sudo usermod -aG developers m_sangeetha11a1
m_sangeetha11a1@cloudshell:~$ newgrp developers
m_sangeetha11a1@cloudshell:~$ groups
developers adm sudo docker m_sangeetha11a1
```

#### 10) su:

```
m_sangeetha11a1@cloudshell:~$ su arun
Password:
arun@cloudshell:/home/m_sangeetha11a1$
```

## CHAPTER 10: COMPRESSION AND ARCHIVING

#### 1) Bzip2:

```
m_sangeetha11a1@cloudshell:~$ touch notes.txt
m_sangeetha11a1@cloudshell:~$ bzip2 notes.txt
m_sangeetha11a1@cloudshell:~$ ls
notes.txt.bz2  README-cloudshell.txt  School  teachers
```

2) Bunzip2:

```
m_sangeetha11a1@cloudshell:~$ bunzip2 notes.txt.bz2
m_sangeetha11a1@cloudshell:~$ ls
notes.txt  README-cloudshell.txt  School  teachers
```

3) Xz:

```
m_sangeetha11a1@cloudshell:~$ xz notes.txt
m_sangeetha11a1@cloudshell:~$ ls
notes.txt.xz  README-cloudshell.txt  School  teachers
```

4) Unxz:

```
m_sangeetha11a1@cloudshell:~$ unxz notes.txt.xz
m_sangeetha11a1@cloudshell:~$ ls
notes.txt  README-cloudshell.txt  School  teachers
```

5) Tar:

```
m_sangeetha11a1@cloudshell:~$ mkdir project
m_sangeetha11a1@cloudshell:~$ touch project/file1.txt
touch project/file2.txt
m_sangeetha11a1@cloudshell:~$ tar -cvf backup.tar project/
project/
project/file2.txt
project/file1.txt
m_sangeetha11a1@cloudshell:~$ ls
backup.tar  notes.txt  project  README-cloudshell.txt  School  teachers
m_sangeetha11a1@cloudshell:~$ tar -tvf backup.tar
drwxrwxr-x m_sangeetha11a1/developers 0 2026-05-15 03:08 project/
-rw-rw-r-- m_sangeetha11a1/developers 0 2026-05-15 03:08 project/file2.txt
-rw-rw-r-- m_sangeetha11a1/developers 0 2026-05-15 03:08 project/file1.txt
```

6) Zip:

```
m_sangeetha11a1@cloudshell:~$ zip files.zip notes.txt
adding: notes.txt (stored 0%)
m_sangeetha11a1@cloudshell:~$ ls
backup.tar  notes.txt  README-cloudshell.txt  teachers
files.zip  project  School
```

7) Unzip:

```
m_sangeetha11a1@cloudshell:~$ unzip files.zip
Archive:  files.zip
```

8) Lzma:

```
m_sangeetha11a1@cloudshell:~$ touch report.txt
m_sangeetha11a1@cloudshell:~$ lzma report.txt
m_sangeetha11a1@cloudshell:~$ ls
backup.tar  notes.txt  README-cloudshell.txt  School
files.zip  project  report.txt.lzma  teachers
```

9) Lzop:

```
m_sangeetha11a1@cloudshell:~$ lzop demo.txt
m_sangeetha11a1@cloudshell:~$ ls
backup.tar  demo.txt.lzo  notes.txt  README-cloudshell.txt  School
demo.txt  files.zip  project  report.txt.lzma  teachers
```

10) Unlzop:

```
demo.txt  files.zip  project  report.txt.lzma  teachers
m_sangeetha11a1@cloudshell:~$ unlzop demo.txt.lzo
```